

ASTR 400B Research Assignment 1

Catherine Snyder

February 2025

1 Research Topic

The topic I would like to focus on for my research assignment is MW/M31 Dark Matter Halo Major Merger Remnant (Number 4 from the list of possible topics).

2 Research Questions

The questions I would like to focus on in my project are:

- Kinematic Properties of the Merged Dark Matter Halos: is there any rotation/what is the evolution of the angular momentum? What is the velocity dispersion profile of the remnant and how does this compare to the original profile of each galaxy?

- What is the contribution of the MW vs. M31 halo particles to the density profile/shape/or kinematics of the merged remnant? Does the contribution from each galaxy have the same spatial orientation/angular momentum direction?

- Specifically comparing how the two galaxies contribute to the angular momentum and how that contribution impacts the evolution of the angular momentum.

3 Alternatives

If there are too many students covering the research topic/research questions, my second choice would be:

- Research Topic: Tidal Transformation of Satellites (M33) (Number 2 from the possible topics)

- Research Questions:

- :Dark Matter Structure: Evolution of M33's dark matter halo - mass loss rates, tidal radius evolution, changes to the internal dark matter profile, changes to the shape of the dark matter distribution.

- :Dark Matter Kinematics: Evolution of the velocity dispersion profile and angular momentum of dark matter particles in M33's (bound/unbound) halo.